



STUDENT PROJECT: CUSTOMER CHURN PREDICTION SYSTEM

Easy-to-Understand Manual for Project
Owners

Developed by: Misbahu Ishaq

Institution: Federal University Dutsin-Ma, Katsina State

Department: Computer Science and IT

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PART 1: UNDERSTANDING YOUR PROJECT



What This Project Is About (In Simple Words)

Imagine you own a telecom company (like MTN, Airtel, Glo). Every month, some customers leave you. We call this "Customer Churn." This project creates a smart computer system that can PREDICT which customers are thinking of leaving, BEFORE they actually leave.

✨ Think of It Like This:

Without This System:

Customer leaves → You find out → Too late to save them

You guess who might leave → Often wrong → Waste money/resources

With This System:

Computer analyzes customer → Says "This customer might leave"

You take action → Customer stays → You save money

Happy customer → More business for you



Key Things You Need to Know:

- It's a PREDICTION tool - Not 100% perfect, but 92.4% accurate
- It gives REASONS - Tells you WHY a customer might leave
- It suggests ACTIONS - Tells you WHAT to do to keep them
- It works with YOUR data - Uses your customer information



PART 2: UNDERSTANDING THE DATA



Your Customer Information File

Your company has a file with customer information. This project uses that file. Here's what the file contains:

20 Pieces of Information About Each Customer:

Information Type	Example	Why It Matters
1. How long with company	24 months	Longer = More loyal
2. Contract type	Monthly vs Yearly	Yearly = Less likely to leave
3. Monthly bill amount	¥5,000	High bill = More complaints
4. Total paid so far	¥120,000	More invested = Harder to leave
5. Age category	Senior citizen?	Different needs
6. Family status	Married? Kids?	Different priorities
7. Phone service	Yes/No	Basic service
8. Internet type	Fiber/DSL/None	Service quality
9. Extra services	Security, backup, etc.	More services = More sticky
10. Payment method	Bank transfer/Cash	Convenience factor

Important Discovery:

Out of 7,043 customers in the sample data:

- 5,174 stayed (73.5%) 
- 1,869 left (26.5%) 

This imbalance means we had to teach the computer to recognize the "leaving" pattern properly.



PART 3: HOW THE SYSTEM WAS BUILT



Step-by-Step Construction Process

STEP 1: Cleaning the Data (Like Washing Vegetables Before Cooking)

Problem: Some customer records had issues. Some had blank total payment fields, some data was in wrong format, and some customer IDs were missing.

Solution: We fixed these issues so the computer could read everything properly.

STEP 2: Understanding Patterns (Like a Detective Looking for Clues)

We created 16 different charts to understand:

- Which contract type loses most customers? → Monthly contracts (43% leave!)
- Which payment method is worst? → Electronic checks (45% leave!)
- How does bill amount affect leaving? → Higher bills = More likely to leave

STEP 3: Building 9 Different "Brain" Models

Model Name	Accuracy	Simple Explanation
XGBoost 🏆	92.4%	Best overall - Like the smartest student
AdaBoost	89.6%	Very good - Learns from mistakes
Random Forest	88.2%	Good - Uses many small decisions
Others	80-88%	Decent but not the best

STEP 4: Creating the "Risk Score" Formula

The system calculates a score from 0-100:

Add Points For Risk Factors:

- New customer (< 3 months): +30 points
- Monthly contract: +25 points
- High bill with fiber internet: +20 points
- Pays by electronic check: +15 points
- Paperless billing: +10 points

Risk Levels:

- 0-40 points → **LOW RISK** (Probably stays)
- 41-70 points → **MEDIUM RISK** (Watch closely)
- 71-100 points → **HIGH RISK** (Act now!)



PART 4: HOW TO USE THE SYSTEM



For Daily Use (Simple Interface)

Option 1: Check Single Customer

CUSTOMER DETAILS FORM:

Customer Name: _____
How long with us: ____ months
Contract: ☐ Monthly ☐ Yearly
Monthly Bill: \$ _____
Internet Type: ☐ Fiber ☐ DSL
Payment: ☐ Bank ☐ Cash ☐ Check

[PREDICT BUTTON]

PREDICTION RESULT:

RISK LEVEL: ● HIGH RISK

SCORE: 85/100

WHY AT RISK:

- ✓ New customer (only 2 months)
- ✓ Monthly contract
- ✓ High monthly bill
- ✓ Pays by check

RECOMMENDED ACTIONS:

1. Call customer within 24 hours
 2. Offer 3-month discount
 3. Suggest yearly contract
 4. Check service quality
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Option 2: Check All Customers at Once

1. Step 1: Export your customer data to Excel
2. Step 2: Upload to the system
3. Step 3: System gives you a report:

CUSTOMER RISK REPORT - January 2026

TOTAL CUSTOMERS: 1,000

-  HIGH RISK: 120 customers (12%)
-  MEDIUM RISK: 300 customers (30%)
-  LOW RISK: 580 customers (58%)

TOP 5 HIGH-RISK CUSTOMERS:

1. John Doe (Score: 95) - Call TODAY
 2. Jane Smith (Score: 90) - Call TODAY
 - ...
-



PART 5: BUSINESS APPLICATION GUIDE



For Customer Service Department

Daily Workflow:

MORNING: Open system → Get "Today's High-Risk List". Call first 10 customers on list. Use provided "retention script". Record outcome in system.

AFTERNOON: Check system for new high-risk. Send personalized emails. Update customer status.

Retention Script Example:

"Hello [Customer Name], this is [Your Name] from [Company]. We noticed you've been with us for [X] months and wanted to check if everything is okay with your service. [IF HIGH RISK] We have a special loyalty offer for you..."



For Sales/Marketing Department

BEFORE CAMPAIGN: Run prediction on all customers. Segment by risk level. Create different offers for each group.

CAMPAIGN OFFERS:

- High Risk: "Stay with us, get 20% off for 6 months"
- Medium Risk: "Renew now, get free month"
- Low Risk: "Refer a friend, get reward"



For Management/Executive

EXECUTIVE DASHBOARD - January 2026



PERFORMANCE METRICS:

- Churn Rate: 3.2% (↓ from 4.5% last month)
 - Predictions Accurate: 92.4%
 - Customers Saved: 156 this month
 - Money Saved: ₦7.8 million
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PART 6: IMPLEMENTATION CHECKLIST

✓ Week 1: Setup Phase

Task	Who Does It
1. Install system on office computer	IT Department
2. Import current customer data	Data Entry
3. Train basic model	Project Team
4. Test with 10 sample customers	Test Team

✓ Month 2: Full Rollout

Task: Full customer base loaded by IT. Daily reporting starts. Integration with CRM. Monthly performance review.



PART 7: COST-BENEFIT ANALYSIS

Your Costs: Training Costs: ₦200,000; Monthly Maintenance: ₦50,000. Total Yearly Cost: ₦800,000.

Your Benefits: Customers Saved: 200 per month; Value per Customer: ₦50,000; Monthly Savings: ₦10,000,000.

NET PROFIT: ₦119,200,000 per year

RETURN ON INVESTMENT: 14,800%



PART 8: MAINTENANCE GUIDE

Monthly: Update customer data, check system accuracy, generate monthly report.

Quarterly: Retrain model with new data, update risk scoring rules, review retention strategies.



PART 9: TROUBLESHOOTING GUIDE

- **Problem:** "System shows all customers as high risk" → Solution: Update with your actual data & retrain.
- **Problem:** "Predictions don't seem right" → Solution: Check data format, contact support.
- **Problem:** "System is slow" → Solution: Restart computer, use batch processing.



PART 10: FUTURE ENHANCEMENTS

Phase 2 (Next 6 Months): Mobile App for field staff, SMS Integration for alerts.

Phase 3 (Next 12 Months): Competitor Price Tracking, Social Media Analysis, Voice Call Analysis.



PART 11: PROJECT DOCUMENTATION

USB CONTENTS: /SYSTEM_FILES (Main program, data), /REPORTS (Docx, PPTX), /SUPPORT (FAQs, Contacts).

Student Information

Project Developer: Misbahu Ishaq

Institution: Federal University Dutsin-Ma, Katsina State

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QUICK START SUMMARY

Day 1-2: Setup. Install system and import data.

Day 3-4: Training. Learn features and create first report.

Day 5: Go Live. Start with high-risk customers.

✨ *GOOD LUCK WITH YOUR CUSTOMER RETENTION JOURNEY!*

For technical issues, contact Misbahu Ishaq.